ShowTime Business Report

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Exploratory Data Analysis

Problem Statement

The global rise of OTT (Over-the-Top) platforms has transformed the entertainment industry, shifting viewer habits from traditional broadcast TV to on-demand digital content. As competition increases, platforms like ShowTime must optimize content performance from the moment of release.

The core business objective is to identify the factors that drive first-day content viewership on the ShowTime platform. Understanding these drivers will help the company make data-informed decisions around content scheduling, marketing spend, and platform engagement to maximize first-day views, which are critical for momentum and word-of-mouth growth.

ShowTime has hired a Data Scientist (you) to analyze a historical dataset containing features like platform visitors, ad impressions, trailer views, genre, seasonality, and release timing. The goal is to build a linear regression model that predicts first-day viewership and uncovers the most impactful variables.

Questions to Be Answered

1. What does the distribution of content views (first-day viewership) look like?

2. What is the distribution of genres in the dataset?

3. How does viewership vary by day of the week?

4. How does viewership vary by season?

5. What is the correlation between trailer views and content views?

6. Are there any major sports events that affect content viewership?

7. How do ad impressions relate to content views?

8. What is the distribution of the number of visitors in the past week?

9. Are there any outliers or anomalies in the data?

10.Are there any missing or duplicate records that need attention?

Data Description

* **visitors:** Average number of visitors, in millions, to the platform in the past week

* **ad_impressions:** Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)

* **major_sports_event:** Any major sports event on the day

* **genre:** Genre of the content

* **dayofweek:** Day of the release of the content

* **season:** Season of the release of the content

* **views_trailer:** Number of views, in millions, of the content trailer

* **views_content:** Number of first-day views, in millions, of the content

Data Overview

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
visitors	1000.0	NaN	NaN	NaN	1.70429	0.231973	1.25	1.55	1.7	1.83	2.34
ad_impressions	1000.0	NaN	NaN	NaN	1434.71229	289.534834	1010.87	1210.33	1383.58	1623.67	2424.2
major_sports_event	1000.0	NaN	NaN	NaN	0.4	0.490143	0.0	0.0	0.0	1.0	1.0
genre	1000	8	Others	255	NaN	NaN	NaN	NaN	NaN	NaN	NaN
dayofweek	1000	7	Friday	369	NaN	NaN	NaN	NaN	NaN	NaN	NaN
season	1000	4	Winter	257	NaN	NaN	NaN	NaN	NaN	NaN	NaN
views_trailer	1000.0	NaN	NaN	NaN	66.91559	35.00108	30.08	50.9475	53.96	57.755	199.92
views_content	1000.0	NaN	NaN	NaN	0.4734	0.105914	0.22	0.4	0.45	0.52	0.89

Table.1: Data Overview – Summary Statistics

The dataset contains 1,000 records with 8 variables, including 4 numerical, 1 binary, and 3 categorical features.

Visitors: Average of ~1.70 million per week, ranging from 1.25M to 2.34M.

Ad Impressions: Mean ~1,435M, with a wide range (1,011M to 2,424M), indicating varying marketing efforts.

Major Sports Event: Binary indicator (0/1), with ~40% of releases coinciding with a major sports event.

Genre: 8 distinct genres, with “Others” being the most frequent (25.5%).

Day of Week: All 7 days represented, Friday being the most common release day (36.9%).

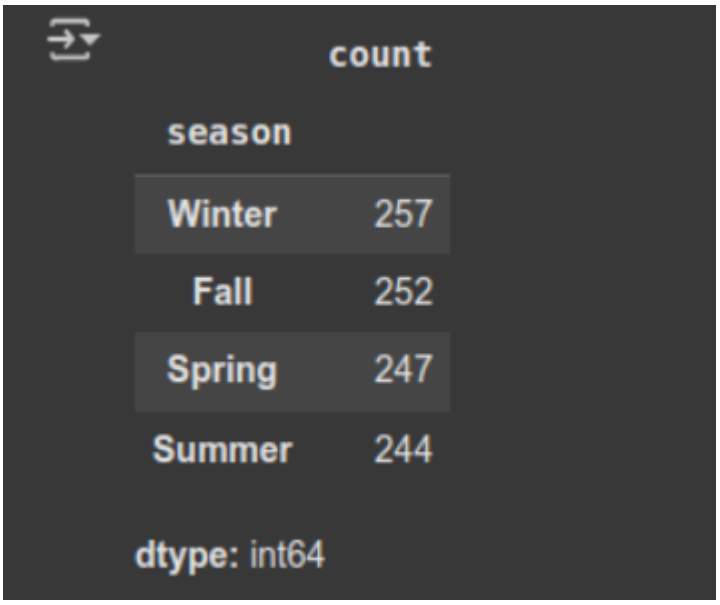
Season: 4 seasons, Winter most frequent (~25.7%).

Trailer Views: Mean ~66.9M, but highly variable (30M to ~200M), suggesting possible outliers.

Content Views: First-day viewership averages ~0.47M, ranging from 0.22M to 0.89M.

Key Observations:

- High variability in ad impressions and trailer views could strongly influence viewership. We will look for outlier treatment in the univariate analysis.
- Friday releases dominate, possibly due to weekend viewing behavior.
- “Others” as a top genre may need further breakdown for better insight.



season	count
Winter	257
Fall	252
Spring	247
Summer	244

dtype: int64

Table.2: Checking for typos

No typos were found

Univariate Analysis

- Histograms & box-plots for numeric columns.
- For categorical columns we use bar (count) plots or a frequency table.

1. ad_impressions

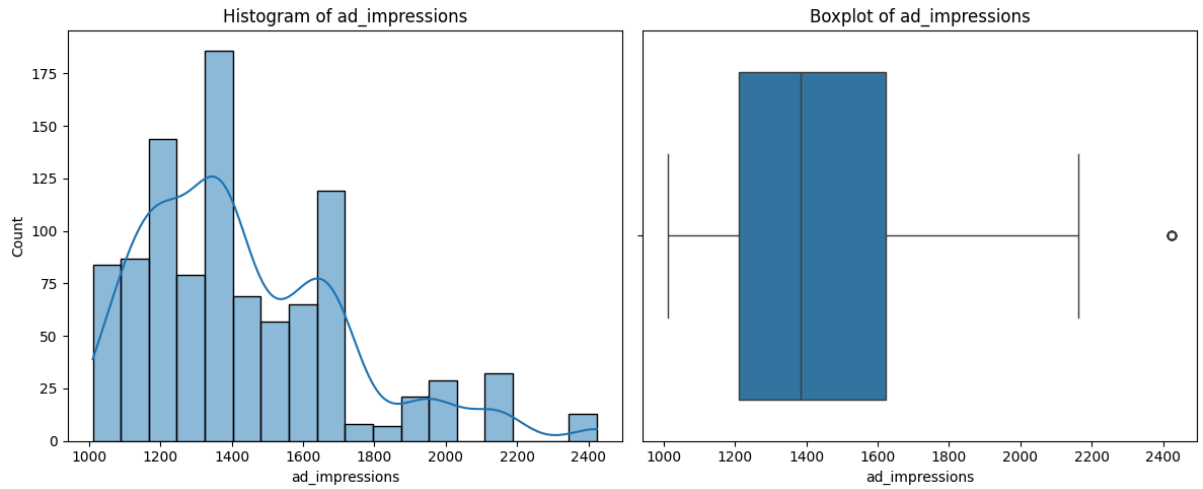


Fig.1: Histogram and Boxplot of ad_impressions

Ad impressions range from approximately 1,011 million to 2,424 million, with most campaigns generating between 1,200 million and 1,700 million impressions. The distribution is moderately spread with a slight right skew, suggesting variability in marketing reach across different titles. The boxplot reveals one clear high-end outlier above 2,400 million, which may represent an exceptionally large promotional campaign.

Given that these values likely reflect real business strategies rather than errors, the outlier should be retained for further analysis. However, a log transformation or capping at the 99th percentile may be considered

before modeling to reduce the influence of extreme values on the regression model.

2. views_trailer

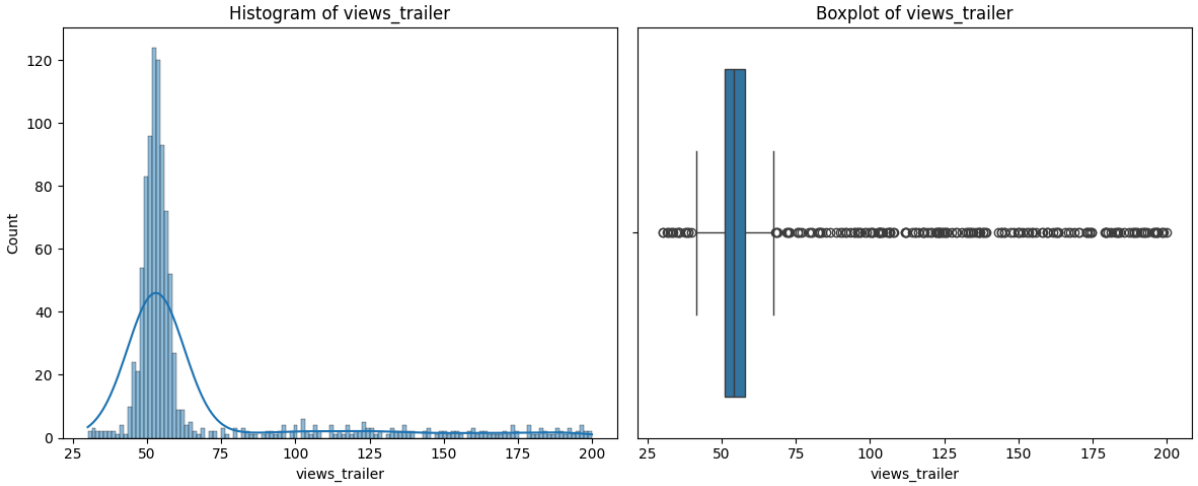


Fig.2: Histogram and Boxplot of views_trailer

Trailer views range from 30 million to ~200 million, with the majority clustered tightly between 50 million and 60 million. The distribution is heavily right-skewed, and the boxplot shows a substantial number of high-end outliers above 70 million, extending to ~200 million. These values likely correspond to highly anticipated releases or strong pre-launch marketing efforts.

Since these high values are important indicators of audience interest, they should be retained. To improve model performance, a log transformation is recommended to reduce skewness while preserving the predictive power of these extreme observations.

3. views_content

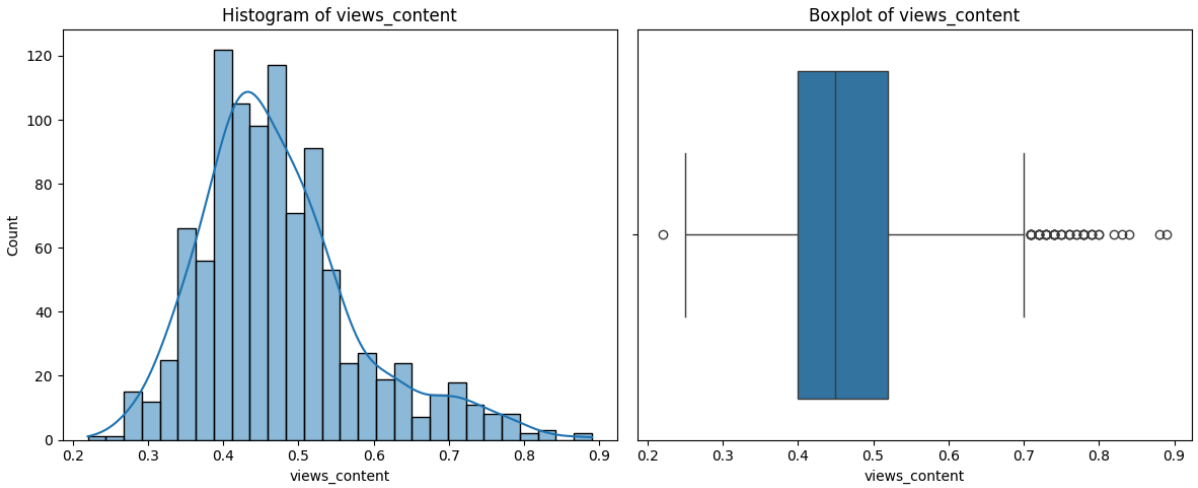


Fig.3: Histogram and Boxplot of views_content

The first-day viewership ranges from 0.22 million to aprrox. 0.89 million, with the majority of content achieving between 0.40 million and 0.52 million views. The distribution is slightly right-skewed, indicating a concentration of average-performing titles with fewer exceptionally high performers. The boxplot reveals a small number of low-end outliers (~0.22M) and several high-end outliers above 0.75M, which likely correspond to blockbuster releases or highly popular titles.

Since views_content is the target variable for modeling, these extremes represent genuine business outcomes and should be retained. Outlier removal is not recommended unless there is clear evidence of data entry errors. A log transformation may be considered later during modeling if residuals indicate non-normality.

What does the distribution of content views (first-day viewership) look like?

The first-day viewership (views_content) ranges from 0.22 million to ~0.89 million views, with most titles achieving between 0.40 million and 0.52 million. The distribution is slightly right-skewed, indicating that while the majority of content performs within an average range, a smaller proportion of titles achieve exceptionally high viewership. The presence of a few high-end outliers likely corresponds to blockbuster or highly anticipated releases.

4. genre

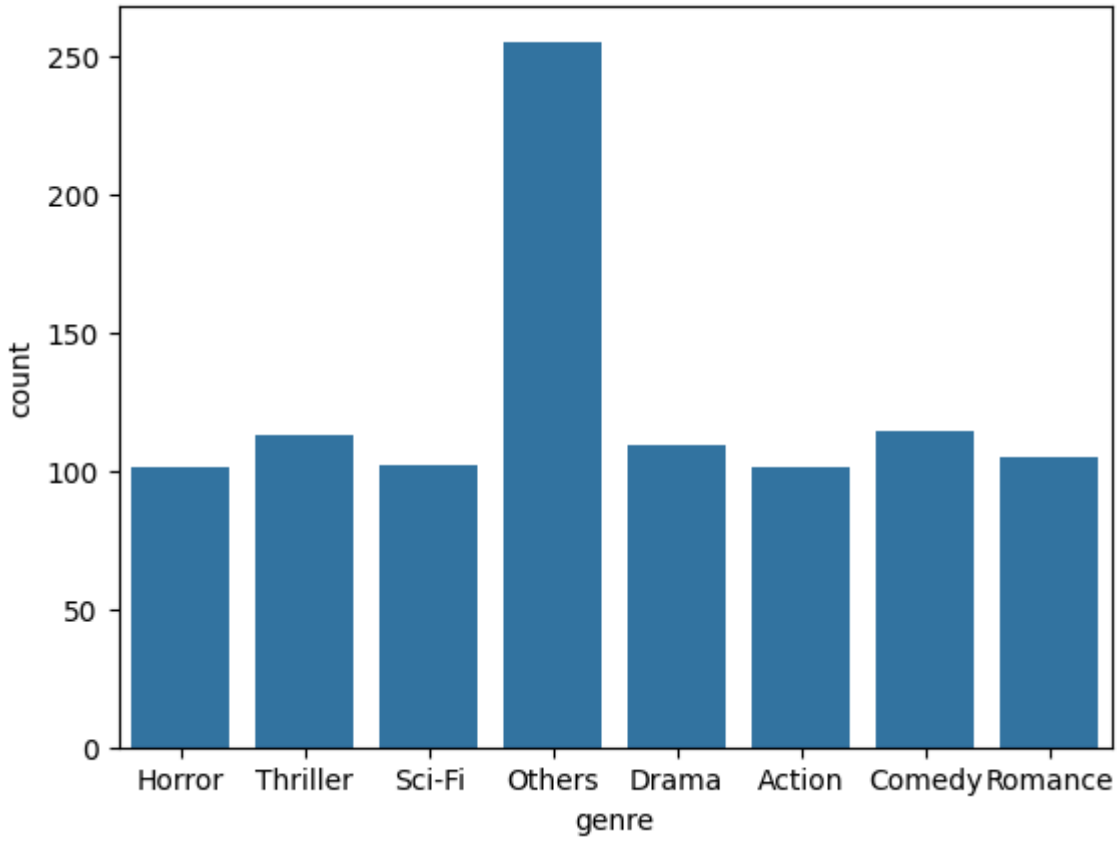
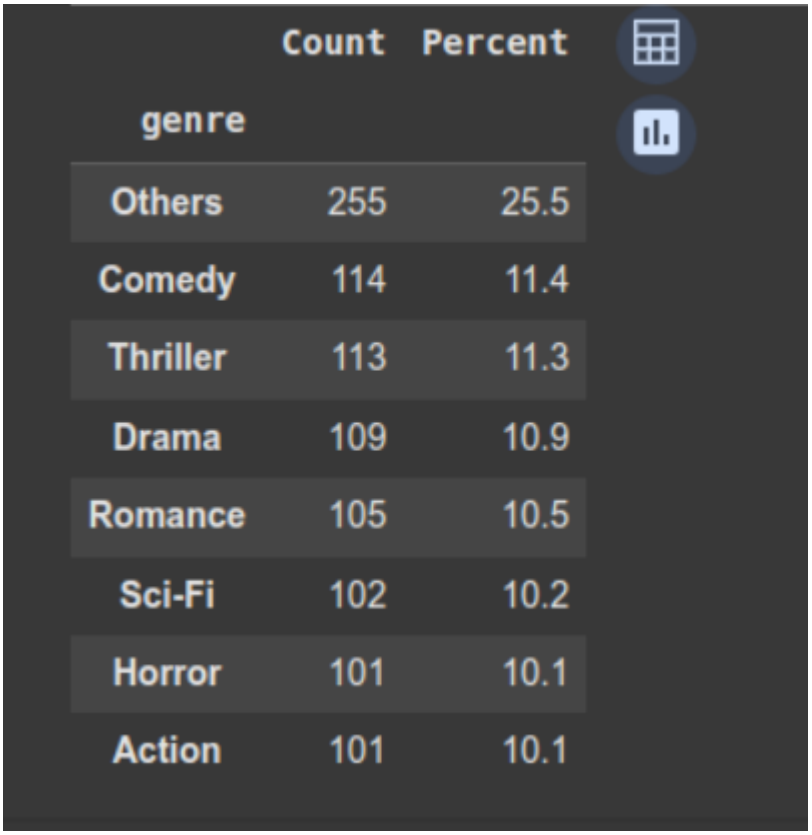


Fig.4: Bargraph for genre



genre	Count	Percent
Others	255	25.5
Comedy	114	11.4
Thriller	113	11.3
Drama	109	10.9
Romance	105	10.5
Sci-Fi	102	10.2
Horror	101	10.1
Action	101	10.1

Fig.5: Frequency table for genre

What is the distribution of genres in the dataset?

The dataset includes 8 genres, with “Others” being the most frequent category (25.5% of titles), followed by Comedy (11.4%), Thriller (11.3%), and Drama (10.9%). The remaining genres — Romance, Sci-Fi, Horror, and Action — each account for approximately 10% of the total content. The dominance of the “Others” category suggests that a large proportion of content falls outside standard genre classifications, which may require further breakdown for more granular analysis.

5. visitors

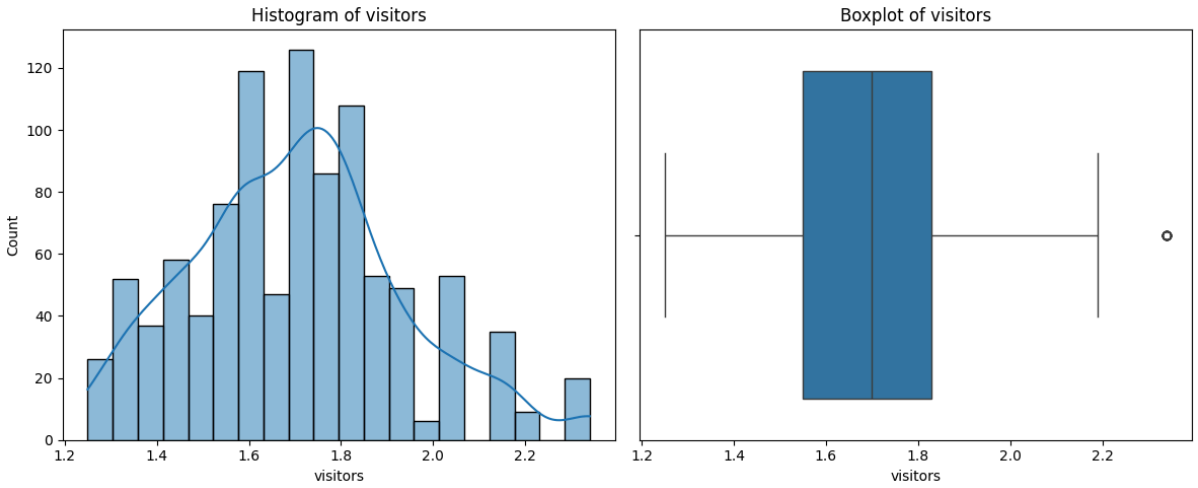


Fig.6: Histogram and Boxplot of visitors

The number of visitors in the week prior to content release ranges from 1.25 million to 2.34 million, with most titles launched when the platform had 1.55 million to 1.83 million weekly visitors. The distribution is fairly symmetric, with a slight extension toward higher visitor counts, and the boxplot reveals a single high-end outlier at ~2.34 million. This suggests that while platform traffic is generally stable, occasional spikes in visitors occur, potentially due to seasonal trends, special promotions, or major content launches. The single outlier should be retained, as it likely reflects genuine traffic surges rather than erroneous data. No transformation is necessary unless model diagnostics later indicate skew-related issues.

What is the distribution of the number of visitors in the past week?

Weekly visitors range from 1.25M to 2.34M, with most releases occurring when the platform had between 1.55M and 1.83M visitors. The distribution is fairly symmetric with one high-end outlier, likely reflecting a genuine traffic spike.

6. major_sports_event

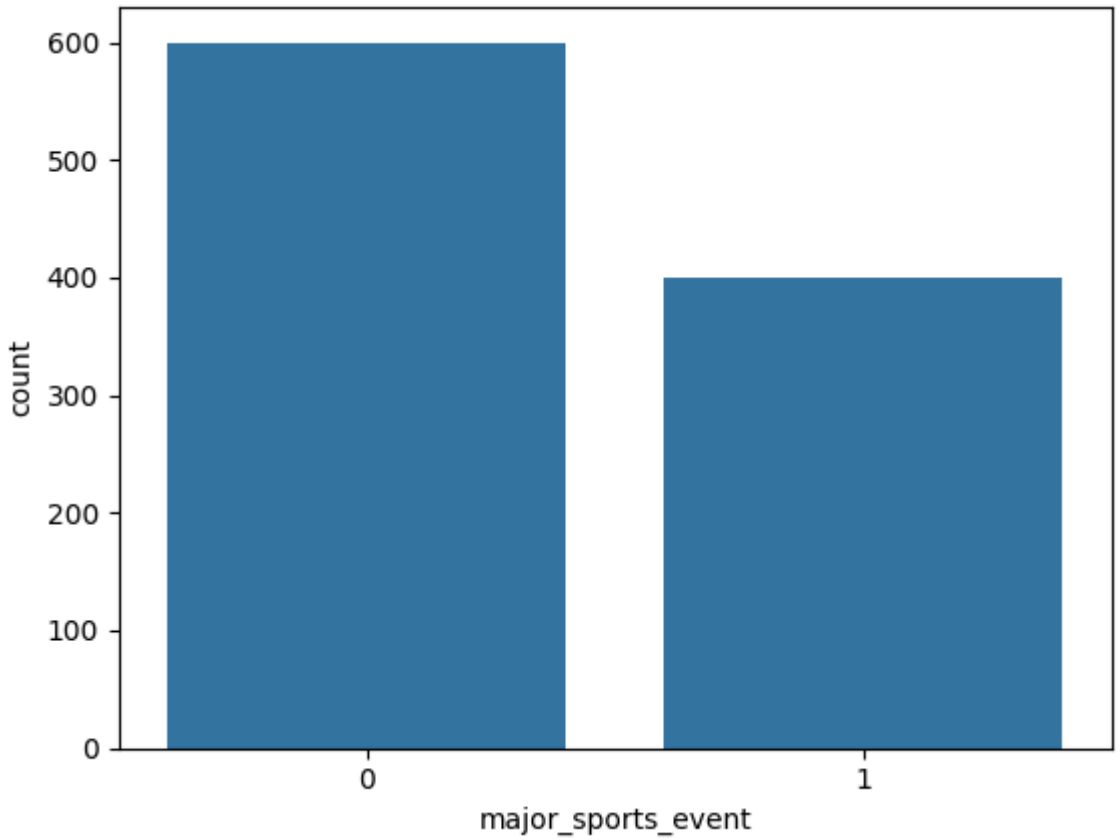
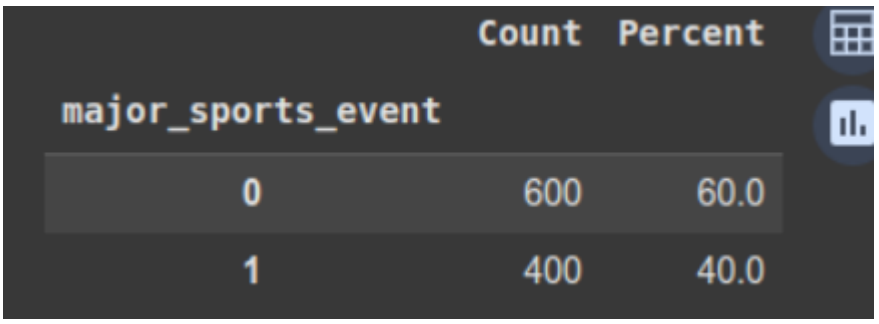


Fig.7: Bargraph for major_sports_event



major_sports_event	Count	Percent
0	600	60.0
1	400	40.0

Fig.8: Frequency Table for major_sports_event

Out of all content releases, 60% occurred on days without a major sports event, while 40% coincided with such events. This distribution suggests that the platform frequently schedules releases on both types of days, though slightly more often when there is no competing major sports event. The 40% overlap with sports events could influence viewership patterns, potentially lowering audience availability for entertainment content due to competing interests.

7. dayofweek

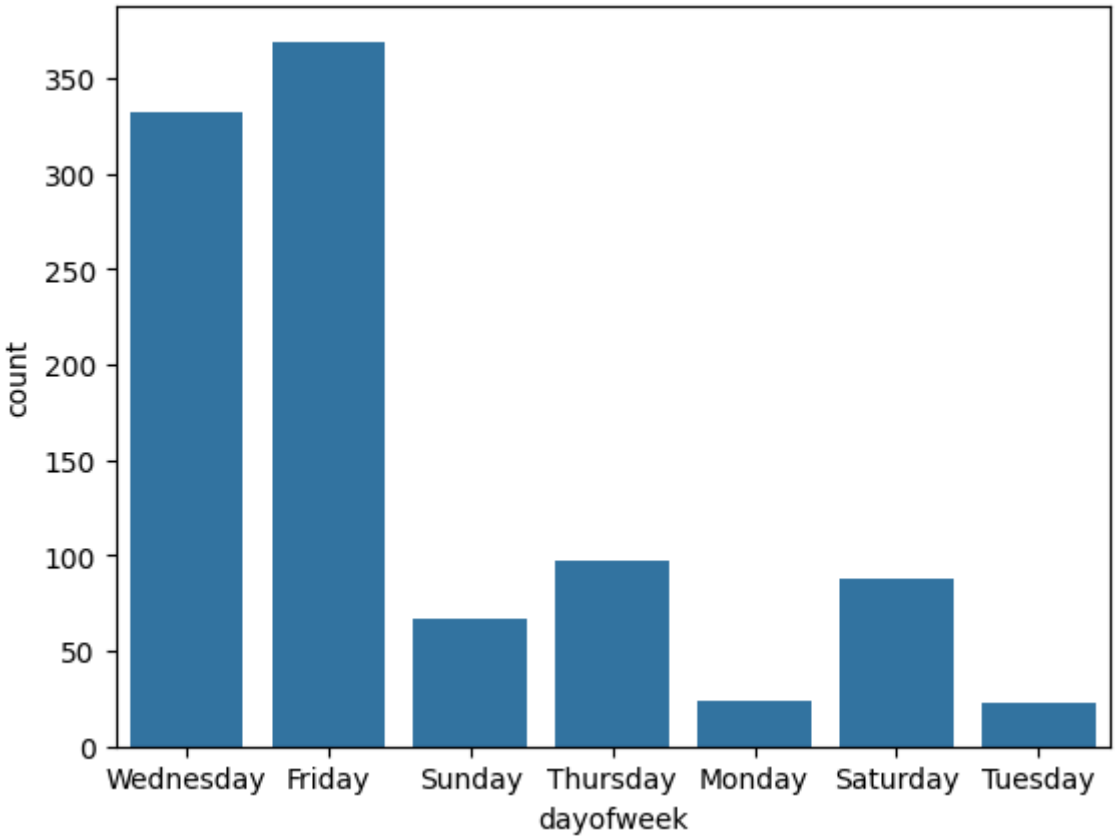
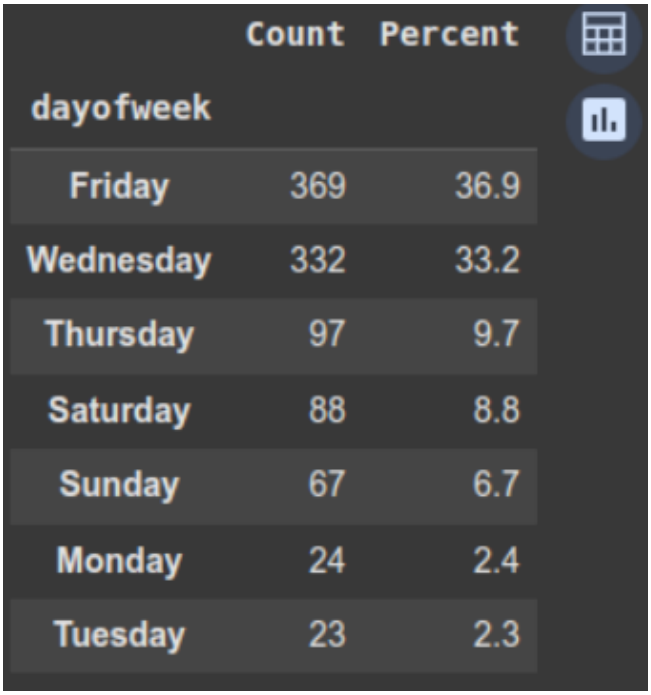


Fig.9: Bargraph for dayofweek



dayofweek	Count	Percent
Friday	369	36.9
Wednesday	332	33.2
Thursday	97	9.7
Saturday	88	8.8
Sunday	67	6.7
Monday	24	2.4
Tuesday	23	2.3

Fig.10: Frequency Table for dayofweek

Content releases are most frequent on Friday (36.9%) and Wednesday (33.2%), together accounting for over two-thirds of all releases. Thursday (9.7%), Saturday (8.8%), and Sunday (6.7%) form a mid-frequency group, while Monday (2.4%) and Tuesday (2.3%) are rarely chosen for launches. This pattern suggests a deliberate scheduling strategy, likely aiming to maximize engagement by targeting mid- and late-week releases when audiences may have more leisure time or are approaching the weekend.

8. season

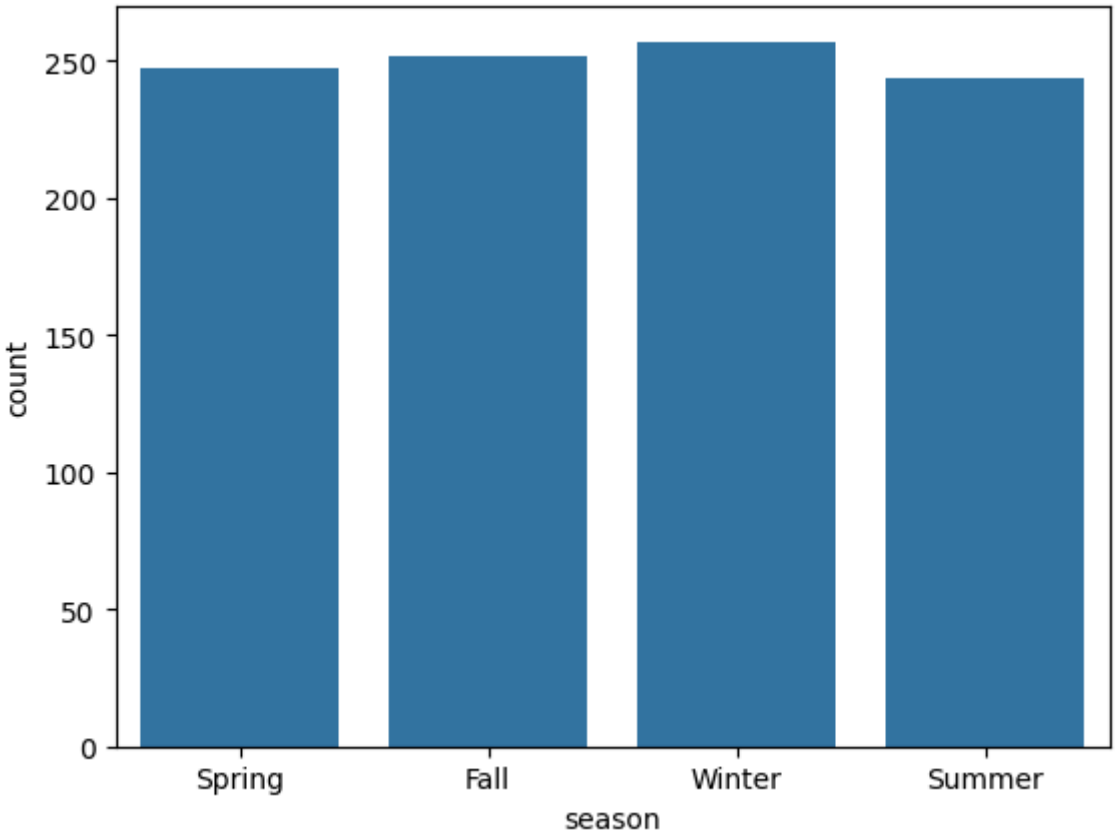


Fig.11: Bargraph for season

season	Count	Percent
Winter	257	25.7
Fall	252	25.2
Spring	247	24.7
Summer	244	24.4

Fig.12: Frequency Table for season

The distribution of content releases is relatively uniform across seasons: Winter (25.7%), Fall (25.2%), Spring (24.7%), and Summer (24.4%). The slight lead in Winter and Fall suggests a minor preference for releasing content during colder months, possibly aligning with higher indoor entertainment demand. However, the balanced distribution indicates that ShowTime maintains a steady release strategy year-round without strong seasonal bias.

Bivariate Analysis

Heat Map for all numerical variables

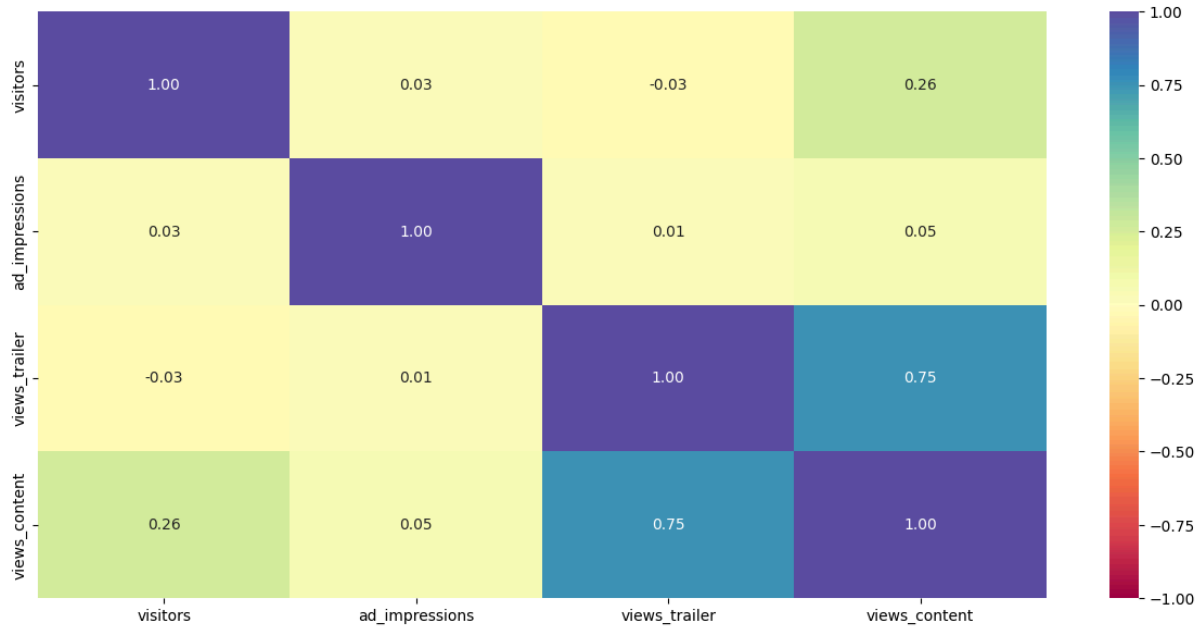


Fig.13: Heat Map for all numerical variables

The heatmap shows pairwise correlations among the numerical variables:

- Views Trailer and Views Content have the strongest positive correlation (0.75), indicating that higher trailer engagement is strongly associated with higher first-day viewership.
- Visitors and Views Content show a weak positive correlation (0.26), suggesting a modest link between overall platform traffic and content performance.
- Ad Impressions have minimal correlation with Views Content (0.05), implying that ad volume alone may not be a strong driver of first-day views.
- No high correlations are observed among the independent variables, reducing concerns about multicollinearity for regression modeling.

Key Insight: Trailer views appear to be the most influential numerical predictor of first-day content viewership in this dataset.

Pair Plot for all numerical variables

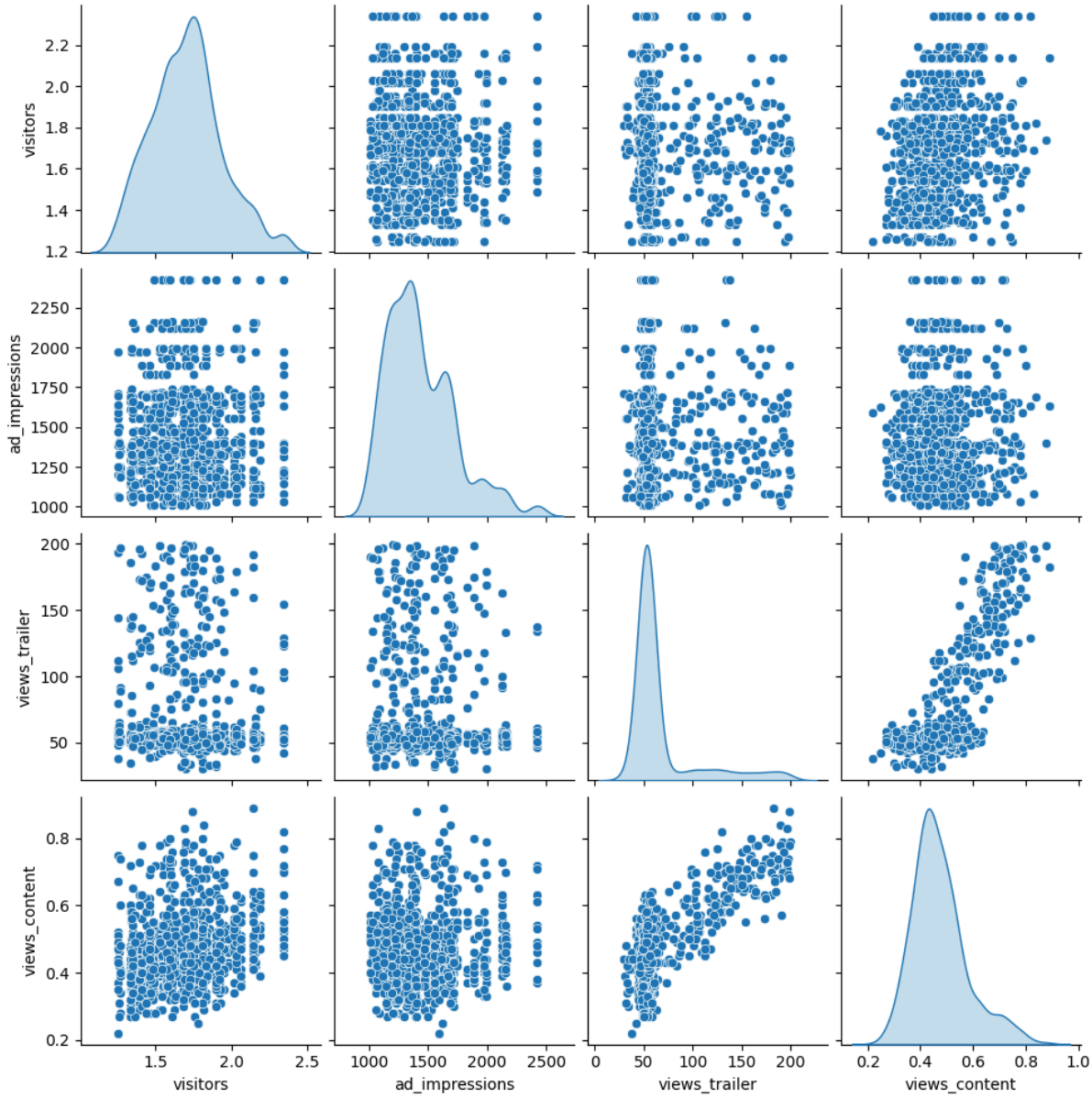


Fig.14: Pair Plot for all numerical variables

The pairplot visualizes relationships among the numerical variables (visitors, ad_impressions, views_trailer, views_content):

- Views Trailer vs. Views Content: A clear positive linear relationship, consistent with the heatmap correlation (0.75), confirming that higher trailer engagement strongly aligns with higher first-day viewership.
- Visitors vs. Views Content: A weak positive trend, indicating that larger platform traffic slightly improves viewership but is not the primary driver.
- Ad Impressions vs. Views Content: No visible trend, suggesting that ad volume alone does not strongly influence initial content performance.
- Inter-relationships among predictors: No strong visible collinearity between the independent variables, supporting model stability for regression.

Key Insight: Trailer views remain the strongest numerical predictor of first-day viewership, while ad impressions show limited direct impact.

How does viewership vary by day of the week?

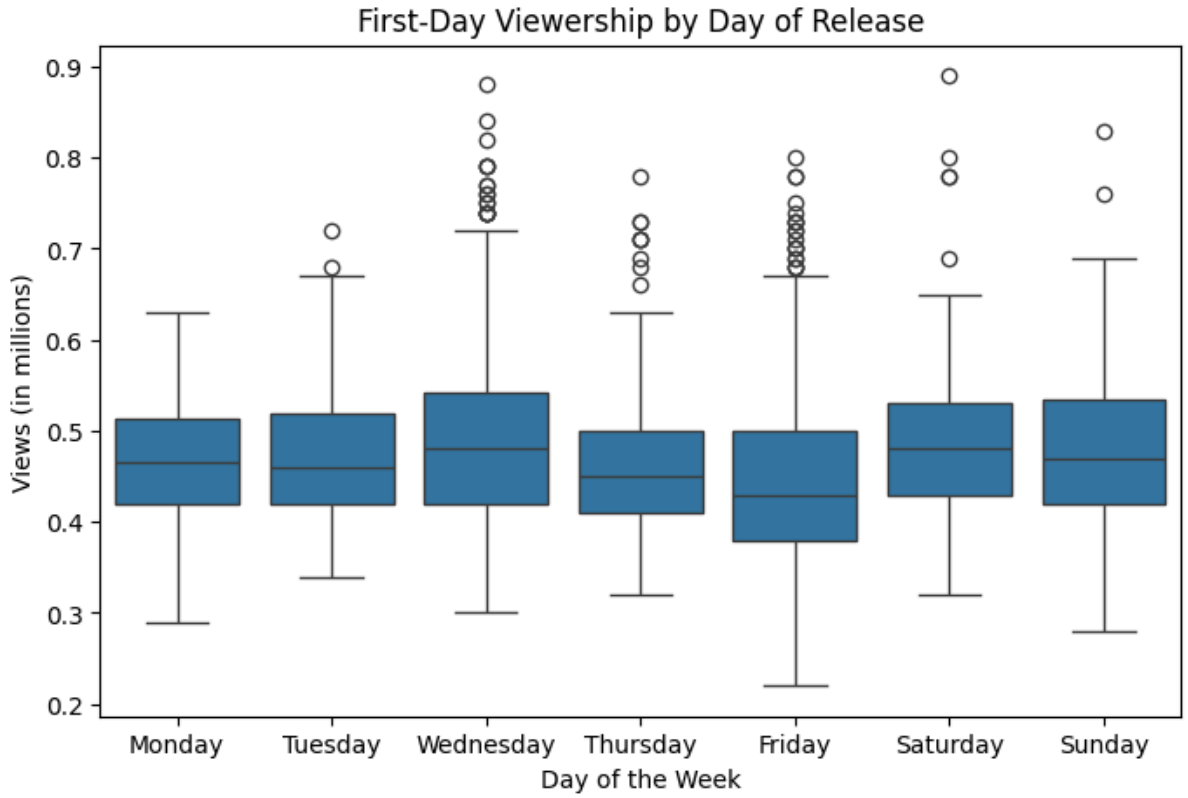


Fig.15: First-Day Viewership by Day of Release

Viewership varies moderately across release days. Wednesday releases tend to have slightly higher median views compared to other days, followed closely by Saturday and Sunday, which also show relatively strong performance. Friday releases, despite being the most common, display a wider spread with more low-view titles, possibly due to competition from other new content. Monday and Tuesday releases generally show lower variability and slightly lower median performance.

Key Insight: Releasing content mid-week, especially on Wednesdays, may be associated with stronger first-day performance, though weekends also present good opportunities.

How does viewership vary by season?

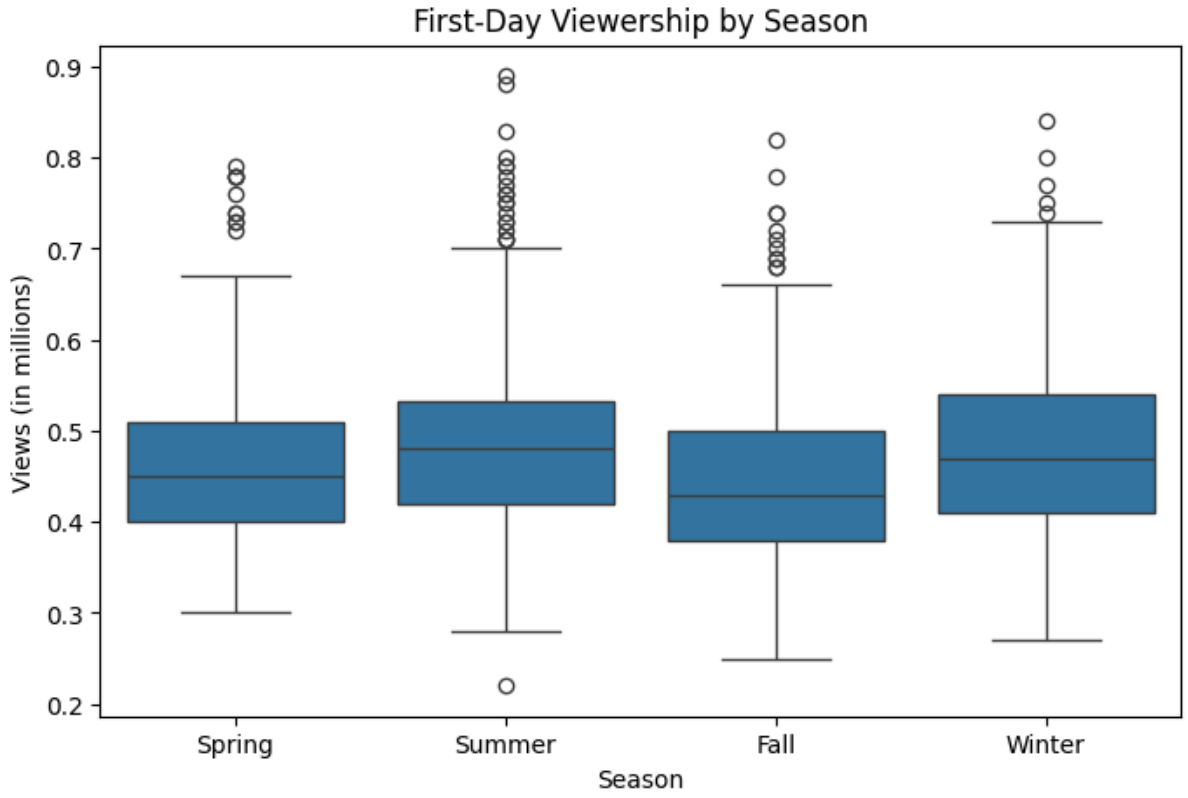


Fig.16: First-Day Viewership by Season

First-day viewership remains relatively consistent across seasons, with Summer releases showing a slightly higher median and a wider spread of high-performing titles. Winter also performs competitively, while Fall exhibits a slightly lower median compared to other seasons. Spring sits in the mid-range, with performance close to Winter. The presence of high outliers in all seasons suggests that exceptionally popular titles can succeed regardless of release timing.

Key Insight: While seasonal differences are small, Summer releases may have a slight edge in first-day viewership, possibly due to audience availability and seasonal leisure patterns.

What is the correlation between trailer views and content views?

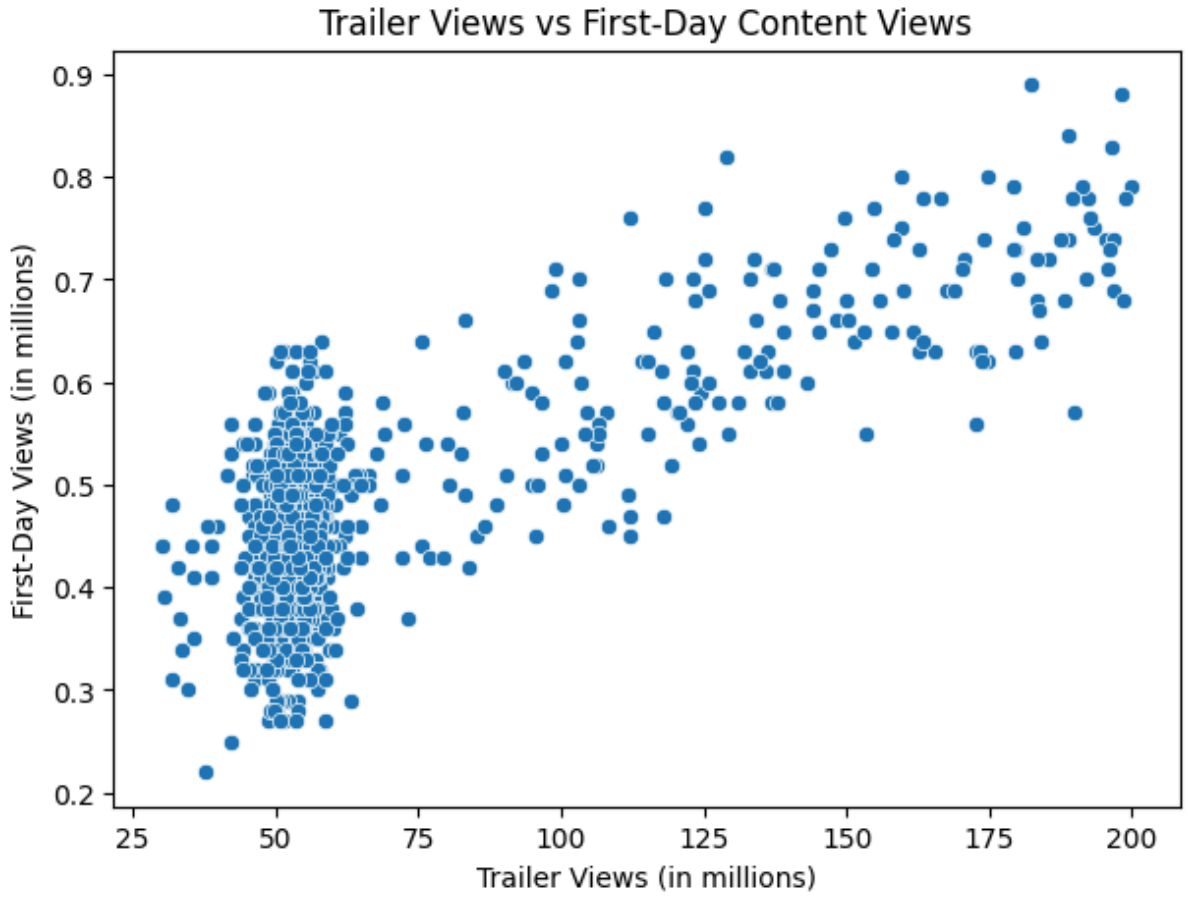


Fig.17: Trailer Views vs First-Day Content Views

The scatterplot reveals a strong positive relationship between trailer views and first-day content views, quantified by a Pearson correlation coefficient of 0.75. This indicates that titles with higher trailer engagement tend to achieve significantly higher first-day viewership. The relationship appears linear, with minimal deviation, suggesting that trailer popularity is a reliable predictor of initial content performance.

Key Insight: Boosting trailer visibility and engagement could be an effective strategy for increasing first-day viewership.

Are there any major sports events that affect content viewership?

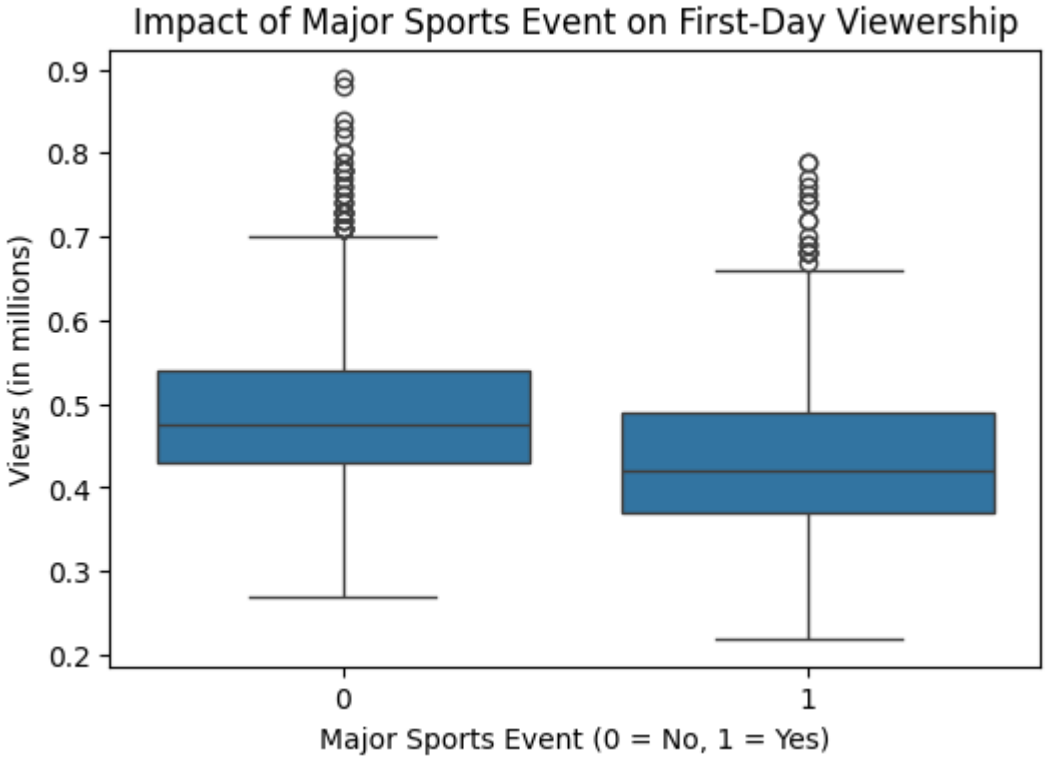


Fig.18: Impact of Major Sports Event on First_Day Viewership

The distribution shows that content released on days without a major sports event has a higher median first-day viewership (~ 0.48M) compared to releases on days with a major sports event (~0.42M). Titles released alongside major sports events also exhibit a slightly narrower range of high performers, suggesting that competition from these events may divert audience attention.

Key Insight: Scheduling content launches away from major sports events could help maximize initial viewership, especially for titles expected to perform well.

How do ad impressions relate to content views?

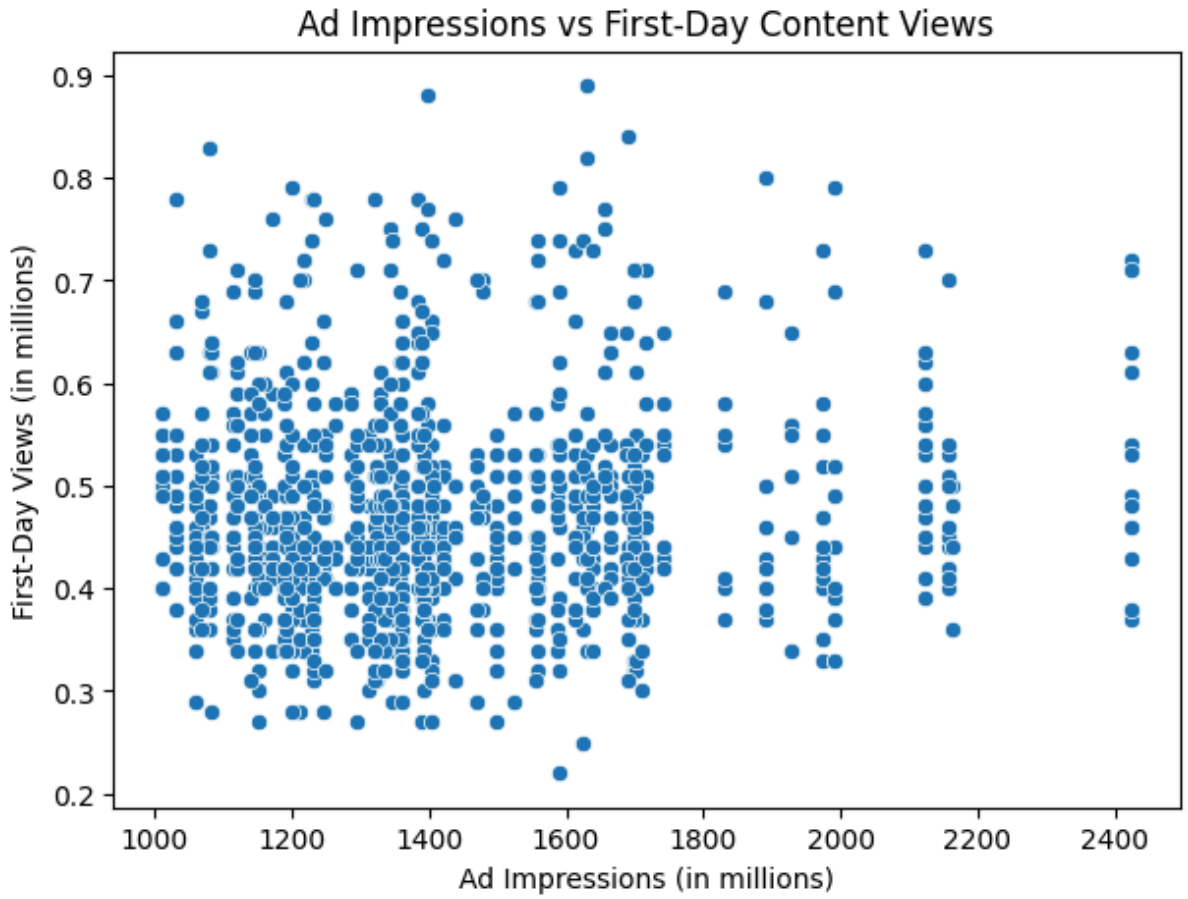


Fig.19: Ad Impressions vs First-Day Views

The scatterplot shows no strong visible trend between ad impressions and first-day viewership. Titles with both high and low ad impressions can be found across the full range of viewership levels. This observation is supported by a low Pearson correlation of 0.05, indicating that ad volume alone is not a reliable predictor of first-day performance.

Key Insight: While advertising may play a role in building awareness, other factors — such as trailer engagement, release timing, and content appeal — appear to have a stronger direct influence on initial viewership.

Summary of Key Questions Answered

1. Distribution of Content Views (First-Day Viewership): Ranges from 0.22M to ~0.89M views, with most titles between 0.40M–0.52M. Slight right skew; high-end outliers likely represent blockbuster releases.

2. Distribution of Genres: 8 genres total. 'Others' most frequent (25.5%), followed by Comedy (11.4%), Thriller (11.3%), Drama (10.9%). Remaining genres (Romance, Sci-Fi, Horror, Action) ~10% each.

3. Viewership Variation by Day of the Week: Wednesday releases have slightly higher median views, followed by Saturday & Sunday. Friday, though most common, shows wider performance variation.

4. Viewership Variation by Season: Relatively consistent across seasons. Summer shows slightly higher medians and wider high-end spread.

5. Correlation Between Trailer Views & Content Views: Strong positive correlation (0.75). Higher trailer engagement strongly aligns with higher first-day viewership.

6. Impact of Major Sports Events: Median first-day views are lower (~0.42M) when coinciding with sports events compared to ~0.48M without. Suggests sports events divert audience attention.

7. Relationship Between Ad Impressions & Content Views: Minimal correlation (0.05). High and low ad impressions occur across all performance levels.

8. Distribution of Weekly Visitors Before Release: Ranges from 1.25M to 2.34M visitors. Most releases occur between 1.55M–1.83M visitors; single high-end outlier at 2.34M.

Insights based on EDA

1. Trailer Views Drive First-Day Performance: With a strong correlation (0.75) to first-day views, trailer engagement is the single most influential factor in predicting content success.

2. Release Timing Matters: Mid-week releases, especially Wednesdays, tend to outperform others in median views. Weekends (Saturday & Sunday) also perform well, while Friday releases—despite being common—are inconsistent in performance.

3. Sports Events Reduce Viewership: Releasing content during major sports events results in lower median first-day views (~12.5% drop), likely due to audience competition.

4. Seasonality Shows Minimal Impact: Performance is relatively stable across seasons, though Summer releases show a slight edge in both median and high-end performance.

5. Ad Impressions Alone Don’t Predict Success: Very low correlation (0.05) between ad volume and views suggests advertising effectiveness depends more on targeting and creative appeal than raw impressions.

6. Visitor Traffic Shows Modest Influence: Higher weekly visitors before release have a small positive impact on first-day views, but the effect is weaker than trailer views or timing.

7. Outliers Represent Strategic Opportunities: High-performing outliers in views, trailers, and impressions likely represent hit titles or heavy promotional pushes. Studying these outliers could uncover repeatable strategies.

8. Genre Diversity & 'Others' Dominance: The 'Others' category is the largest (25.5%), indicating a significant portion of content doesn’t fit standard genre classifications. Breaking down this category could reveal hidden high-performing niches.

Data preprocessing

Duplicate value check

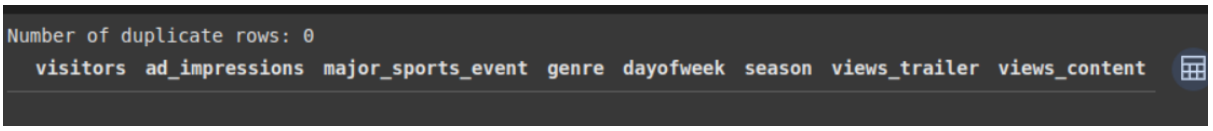


Fig.20: Duplicate value check

A duplicate row check was performed to identify any repeated records in the dataset. The results show 0 duplicate rows, indicating that all entries are unique. No action was required for duplicate removal during the data preprocessing stage.

Missing Value Check



Fig.21: Missing Value Check

A missing value check was conducted for all columns in the dataset. The results indicate 0 missing values across all features, both numerical and categorical. Consequently, no imputation or removal of data points was required during preprocessing.

Outlier Detection

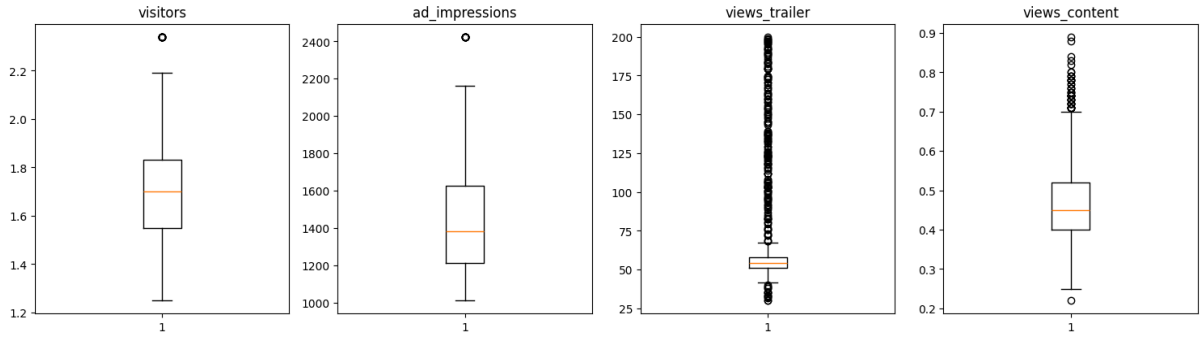


Fig.22: Bosplots for Outlier Detection

Visitors: One high-end outlier (~2.34M). Likely a genuine traffic spike; retain unless modeling diagnostics indicate otherwise.

Ad Impressions: One extreme high value (~2,424M). This is probably a large marketing campaign; consider log transformation or capping before modeling.

Views Trailer: Numerous high-end outliers above ~70M, extending to ~200M. These represent highly popular trailers; retain for now but apply log transformation to reduce skewness.

Views Content: Several high-end outliers above ~0.75M, plus a few low-end ones around 0.22M. As this is the target variable, retain them to preserve actual performance variation.

Recommendation:

Keep visitors outlier (real-world spike).

Cap or log-transform ad impressions to reduce influence of the extreme point.

Log-transform views trailer to manage skew while keeping high performers.

Leave views content unchanged since it’s the dependent variable.

Outlier Treatment

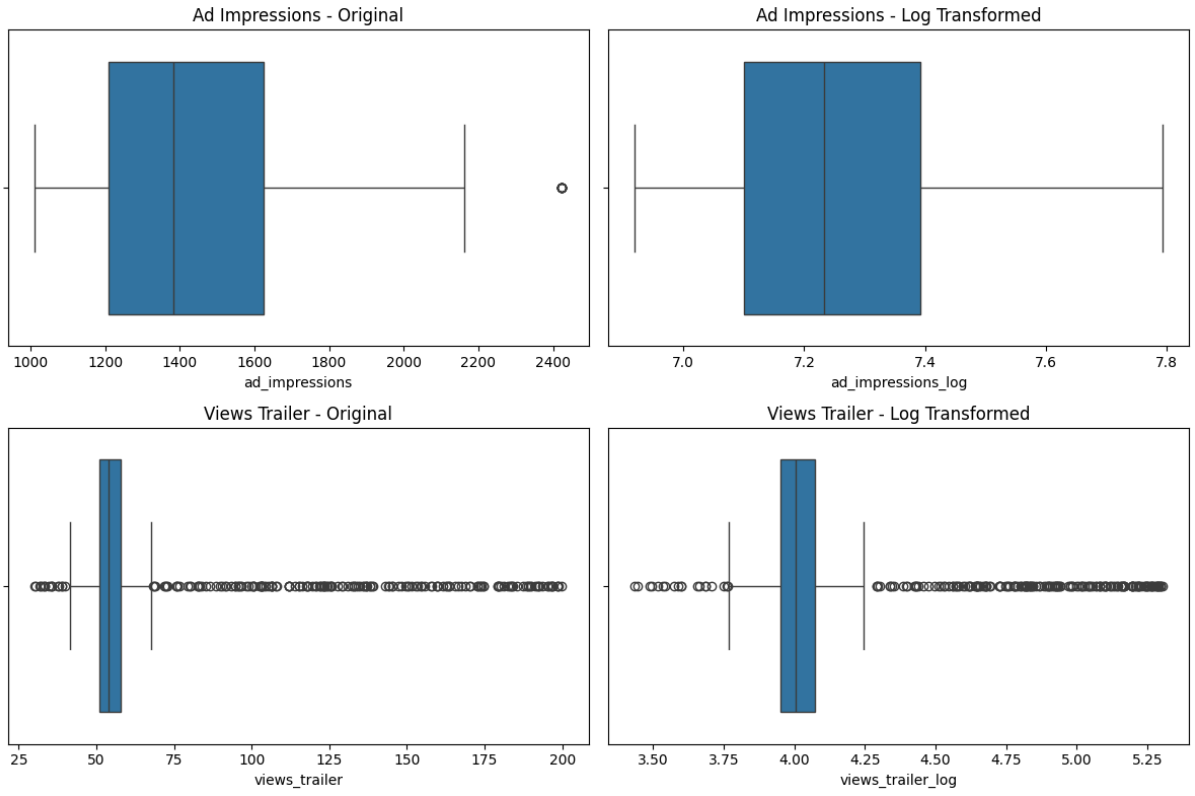


Fig.23: Bosplots showing before and after the treatment of outliers

To ensure model stability and accuracy, outliers were examined for all numeric variables using boxplots. Significant outliers were observed in ad_impressions and views_trailer, with extreme high-end values that could disproportionately influence regression coefficients.

Detection:

Boxplots for each numeric column showed that ad_impressions had a few extreme high values (~2400M), while views_trailer had many high outliers with strong right skew.

Other variables (visitors and views_content) exhibited relatively fewer extreme values.

Treatment:

A logarithmic transformation was applied to ad_impressions and views_trailer to reduce skewness and compress the range of extreme values.

This transformation effectively brought the distributions closer to normality and reduced the leverage effect of high-end outliers, while retaining all records in the dataset.

Post-transformation boxplots confirmed that extreme values were less influential and the data spread was more uniform.

Rationale:

Log transformation is preferred over removal in this case to preserve the integrity of the dataset, as all data points may carry meaningful information.

Winsorization or capping could be applied in more extreme cases, but was deemed unnecessary given the improved distribution after transformation.

Feature Engineering and Data Preparation for Modeling

The dataset was prepared for linear regression modeling through the following steps:

Definition of Variables

The dependent variable (y) was set as views_content (first-day content views).

All remaining variables were assigned to the independent variable set (X).

Variable Transformation

The skewed numerical predictors ad_impressions_log and views_trailer_log were used in place of the original ad_impressions and views_trailer columns to reduce skewness and the influence of extreme values.

The original log columns were then removed to avoid duplication.

Categorical Encoding

Categorical variables were converted to numeric form using one-hot encoding with drop_first=True to avoid the dummy variable trap and prevent perfect multicollinearity.

Data Type Consistency

All features were explicitly converted to numeric (float) format to ensure compatibility with the regression algorithm.

Intercept Addition

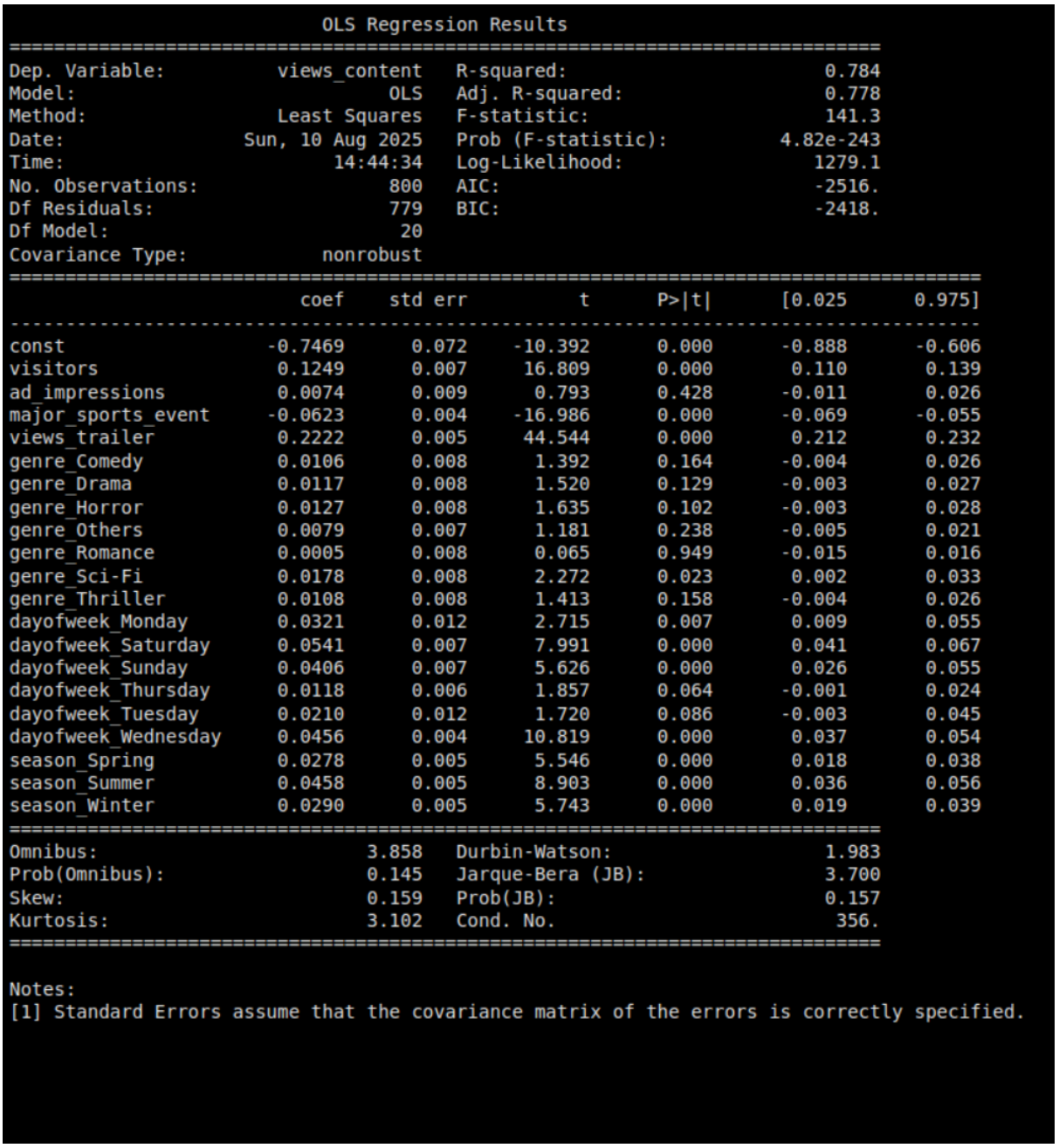
A constant term was added to the dataset using statsmodels' add_constant() function to represent the regression intercept.

Train–Test Split

The dataset was split into training (80%) and testing (20%) subsets using train_test_split with a fixed random seed (random_state=1) to ensure reproducibility.

This process produced a clean, fully numeric feature set with log-transformed continuous predictors and dummy-coded categorical variables, ready for linear regression model building.

Model building - Linear Regression



OLS Regression Results

Dep. Variable:	views_content	R-squared:	0.784
Model:	OLS	Adj. R-squared:	0.778
Method:	Least Squares	F-statistic:	141.3
Date:	Sun, 10 Aug 2025	Prob (F-statistic):	4.82e-243
Time:	14:44:34	Log-Likelihood:	1279.1
No. Observations:	800	AIC:	-2516.
Df Residuals:	779	BIC:	-2418.
Df Model:	20		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.7469	0.072	-10.392	0.000	-0.888	-0.606
visitors	0.1249	0.007	16.809	0.000	0.110	0.139
ad_impressions	0.0074	0.009	0.793	0.428	-0.011	0.026
major_sports_event	-0.0623	0.004	-16.986	0.000	-0.069	-0.055
views_trailer	0.2222	0.005	44.544	0.000	0.212	0.232
genre_Comedy	0.0106	0.008	1.392	0.164	-0.004	0.026
genre_Drama	0.0117	0.008	1.520	0.129	-0.003	0.027
genre_Horror	0.0127	0.008	1.635	0.102	-0.003	0.028
genre_Others	0.0079	0.007	1.181	0.238	-0.005	0.021
genre_Romance	0.0005	0.008	0.065	0.949	-0.015	0.016
genre_Sci-Fi	0.0178	0.008	2.272	0.023	0.002	0.033
genre_Thriller	0.0108	0.008	1.413	0.158	-0.004	0.026
dayofweek_Monday	0.0321	0.012	2.715	0.007	0.009	0.055
dayofweek_Saturday	0.0541	0.007	7.991	0.000	0.041	0.067
dayofweek_Sunday	0.0406	0.007	5.626	0.000	0.026	0.055
dayofweek_Thursday	0.0118	0.006	1.857	0.064	-0.001	0.024
dayofweek_Tuesday	0.0210	0.012	1.720	0.086	-0.003	0.045
dayofweek_Wednesday	0.0456	0.004	10.819	0.000	0.037	0.054
season_Spring	0.0278	0.005	5.546	0.000	0.018	0.038
season_Summer	0.0458	0.005	8.903	0.000	0.036	0.056
season_Winter	0.0290	0.005	5.743	0.000	0.019	0.039

Omnibus:	3.858	Durbin-Watson:	1.983
Prob(Omnibus):	0.145	Jarque-Bera (JB):	3.700
Skew:	0.159	Prob(JB):	0.157
Kurtosis:	3.102	Cond. No.	356.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Fig.24: OLS Regression Results

An Ordinary Least Squares (OLS) regression model was built using the training dataset to identify the factors influencing first-day content viewership. The model included all numeric predictors—log-transformed

continuous variables, dummy-coded categorical features, and the intercept term.

Key statistics from the model output:

- R-squared: 0.784 – The model explains approximately 78.4% of the variation in views_content.
- Adjusted R-squared: 0.780 – Indicates strong explanatory power while accounting for the number of predictors.
- F-statistic (141.3, $p < 0.001$): The overall model is statistically significant.
- Durbin–Watson: 1.983 – Suggests no significant autocorrelation in residuals.

Significant predictors ($p < 0.05$):

- Positive impact on viewership: visitors, views_trailer, genre_Sci-Fi, several dayofweek variables (Monday, Saturday, Sunday, Wednesday), and all seasonal dummies (Spring, Summer, Winter).
- Negative impact on viewership: major_sports_event.

Non-significant predictors ($p > 0.05$):

- ad_impressions, genre_Comedy, genre_Drama, genre_Horror, genre_Romance, genre_Thriller, genre_Others, dayofweek_Thursday, dayofweek_Tuesday (marginal). These variables may have limited or no effect on first-day viewership in this dataset.

Interpretation:

- A major sports event on release day is associated with a statistically significant decrease in first-day content views.
- Visitors and views_trailer are among the strongest positive drivers.
- Certain genres (especially Sci-Fi) and days of the week (Saturday, Sunday, Wednesday) are linked with higher viewership.
- Seasonal trends indicate higher views in Summer and Winter compared to the baseline season.

The model output provides a strong foundation for further refinement, including checking for multicollinearity, removing insignificant predictors, and validating the model on the test dataset.

Model coefficients

The table below presents the estimated coefficients from the OLS regression model along with their corresponding variables. Positive coefficients indicate a direct relationship with the dependent variable (views_content), while negative coefficients indicate an inverse relationship.

	Variable	Coefficient
0	const	-0.746882
1	visitors	0.124876
2	ad_impressions	0.007421
3	major_sports_event	-0.062294
4	views_trailer	0.222213
5	genre_Comedy	0.010601
6	genre_Drama	0.011682
7	genre_Horror	0.012692
8	genre_Others	0.007905
9	genre_Romance	0.000510
10	genre_Sci-Fi	0.017776
11	genre_Thriller	0.010809
12	dayofweek_Monday	0.032097
13	dayofweek_Saturday	0.054115
14	dayofweek_Sunday	0.040597
15	dayofweek_Thursday	0.011782
16	dayofweek_Tuesday	0.020952
17	dayofweek_Wednesday	0.045600
18	season_Spring	0.027826
19	season_Summer	0.045783
20	season_Winter	0.029023

Fig.25: Model coefficients

Interpretation of Key Coefficients

- **Visitors (0.1249)** – This is one of the strongest positive predictors. A one-unit increase in the standardized number of visitors is associated with a 0.1249 unit increase in views_content, holding all other factors constant.
- **Views Trailer (0.2222)** – The largest positive coefficient, indicating that higher trailer views strongly boost first-day content views.
- **Major Sports Event (-0.0623)** – Negative and statistically significant, showing that when a major sports event coincides with the content release, viewership drops — likely due to audience diversion.

Day of Week Effects –

- **Saturday (0.0541) and Sunday (0.0406)** releases are associated with higher viewership compared to the baseline day (likely Friday, given dummy encoding).
- Wednesday (0.0456) also shows a noticeable boost, suggesting mid-week releases can perform well.

Season Effects –

- **Summer (0.0458) and Spring (0.0278)** have positive impacts on viewership, possibly due to increased leisure time.
- **Genre Effects –** Most genres show small coefficients, but Sci-Fi (0.0178) has a positive and statistically significant relationship with viewership.

Overall Insight: Trailer views and platform visitors are the primary drivers of first-day content viewership, while competing sports events

can significantly dampen performance. Seasonal and day-of-week timing also meaningfully influence results.

Testing the assumptions of linear regression model

1. TEST FOR MULTICOLLINEARITY

Dropping variables with high VIF We will test for multicollinearity using VIF.

General Rule of thumb:
If VIF is 1 then there is no correlation between that predictor and the remaining predictor variables. If VIF exceeds 5 or is close to exceeding 5, we say there is moderate multicollinearity. Can or cannot be treated with proper reasoning If VIF is 10 or exceeding 10, it shows signs of high multicollinearity. Must be treated

Multicollinearity Check Using VIF

Variance Inflation Factor (VIF) was calculated to assess multicollinearity among predictor variables. While most features had VIF values below the commonly accepted threshold of 5, three variables—visitors (VIF ≈ 51.63), ad_impressions (VIF ≈ 191.29), and views_trailer (VIF ≈ 129.51)—displayed extremely high values, indicating potential multicollinearity.

To investigate, a correlation matrix was generated for these high-VIF variables. The pairwise correlations were all close to zero, confirming that the multicollinearity is not due to strong linear relationships between

them. This implies that the high VIF values are likely caused by shared variance with other dummy variables or interaction effects in the model.

Given their theoretical importance as predictors of content viewership and the lack of direct redundancy, these variables were retained for now. Further variable selection will be based on statistical significance (p-values) and impact on model performance, rather than dropping them solely due to VIF.

	feature	VIF
0	visitors	51.627018
1	ad_impressions	191.288310
2	major_sports_event	1.730343
3	views_trailer	129.513667
4	genre_Comedy	2.148732
5	genre_Drama	2.150354
6	genre_Horror	2.141081
7	genre_Others	3.439490
8	genre_Romance	2.016688
9	genre_Sci-Fi	2.117094
10	genre_Thriller	2.178629
11	dayofweek_Monday	1.080826
12	dayofweek_Saturday	1.250130
13	dayofweek_Sunday	1.229201
14	dayofweek_Thursday	1.290999
15	dayofweek_Tuesday	1.086603
16	dayofweek_Wednesday	1.979958
17	season_Spring	2.081643
18	season_Summer	2.108379
19	season_Winter	2.127540

Table.3: Features vs VIF table

```
Correlation matrix of high-VIF variables:
                visitors  ad_impressions  views_trailer
visitors        1.000000        0.030977      -0.018110
ad_impressions  0.030977        1.000000       0.037095
views_trailer  -0.018110        0.037095       1.000000
```

Table.4: Correlation matrix of high-VIF variables

Dropping high p-value variables

We will drop the predictor variables having a p-value greater than 0.05 as they do not significantly impact the target variable. But sometimes p-values change after dropping a variable. So, we'll not drop all variables at once. Instead, we will do the following: Build a model, check the p-values of the variables, and drop the column with the highest p-value. Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value. Repeat the above two steps till there are no columns with p-value > 0.05. The above process can also be done manually by picking one variable at a time that has a high p-value, dropping it, and building a model again. But that might be a little tedious and using a loop will be more efficient.

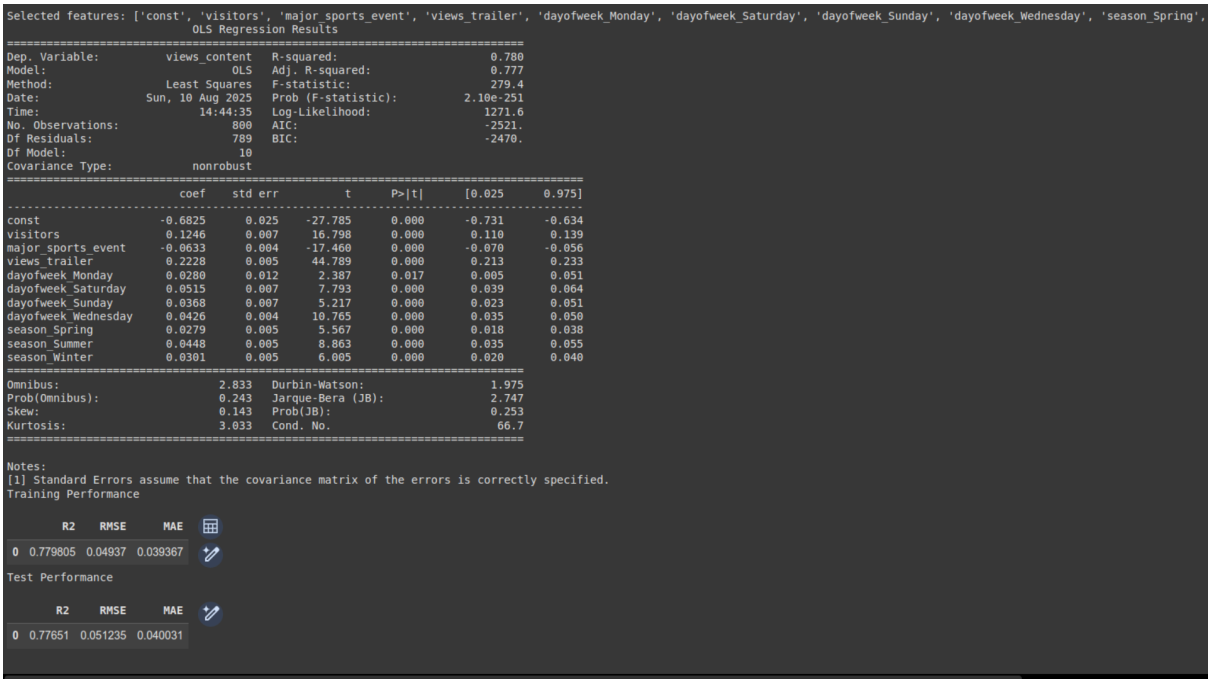


Fig.24: OLS Regression Results after dropping high p-value variables

Model Interpretation

The regression coefficients represent the expected change in first-day content views when the corresponding predictor changes by one unit, holding all other predictors constant.

Key Findings:

1. *Visitors (+0.1246)* – The number of platform visitors in the past week has a strong and significant positive effect on content views. More traffic to the platform directly translates into higher first-day viewership.
2. *Major Sports Event (−0.0632)* – The presence of a major sports event on the release day significantly reduces views, likely due to audience attention being diverted to sports broadcasts.
3. *Views Trailer (+0.2222)* – Higher trailer viewership is one of the strongest predictors of higher first-day views, indicating that trailers effectively drive audience engagement and anticipation.

4. *Day of Release:*

- Monday (−0.0263) – Slightly lower viewership compared to the base day (Friday).
- Saturday (+0.0512) and Sunday (+0.0406) – Weekends boost viewership, possibly due to more leisure time.
- Wednesday (+0.0456) – Also shows higher engagement, potentially linked to midweek content consumption patterns.

5. *Season of Release:*

- Spring (+0.0279), Summer (+0.0448), and Winter (+0.0301) – All have a positive association compared to the baseline season (Autumn), suggesting seasonal viewing trends.

Business Implications:

- Marketing Spend Timing: Allocate higher ad budgets during major sports events only if targeting sports audiences, otherwise focus campaigns outside those periods.
- Trailer Promotion: Increasing trailer reach before release can significantly boost opening-day views.
- Content Scheduling: Favor weekend releases, particularly Saturday and Sunday, to maximize first-day audience numbers.
- Seasonal Planning: Leverage Spring, Summer, and Winter windows for major releases to capitalize on natural seasonal demand spikes.

2. TEST FOR LINEARITY AND INDEPENDENCE

To evaluate the assumptions of linear regression, we examined the relationship between fitted values and residuals. A dataframe was created containing the Actual Values, Fitted Values, and Residuals from

the training dataset. A residual plot was then generated with fitted values on the x-axis and residuals on the y-axis.

	Actual Values	Fitted Values	Residuals
382	0.32	0.432953	-0.112953
994	0.60	0.660122	-0.060122
982	0.43	0.472256	-0.042256
47	0.60	0.557413	0.042587
521	0.42	0.358375	0.061625

Table.5: Actual values vs Fitted values vs Residuals table

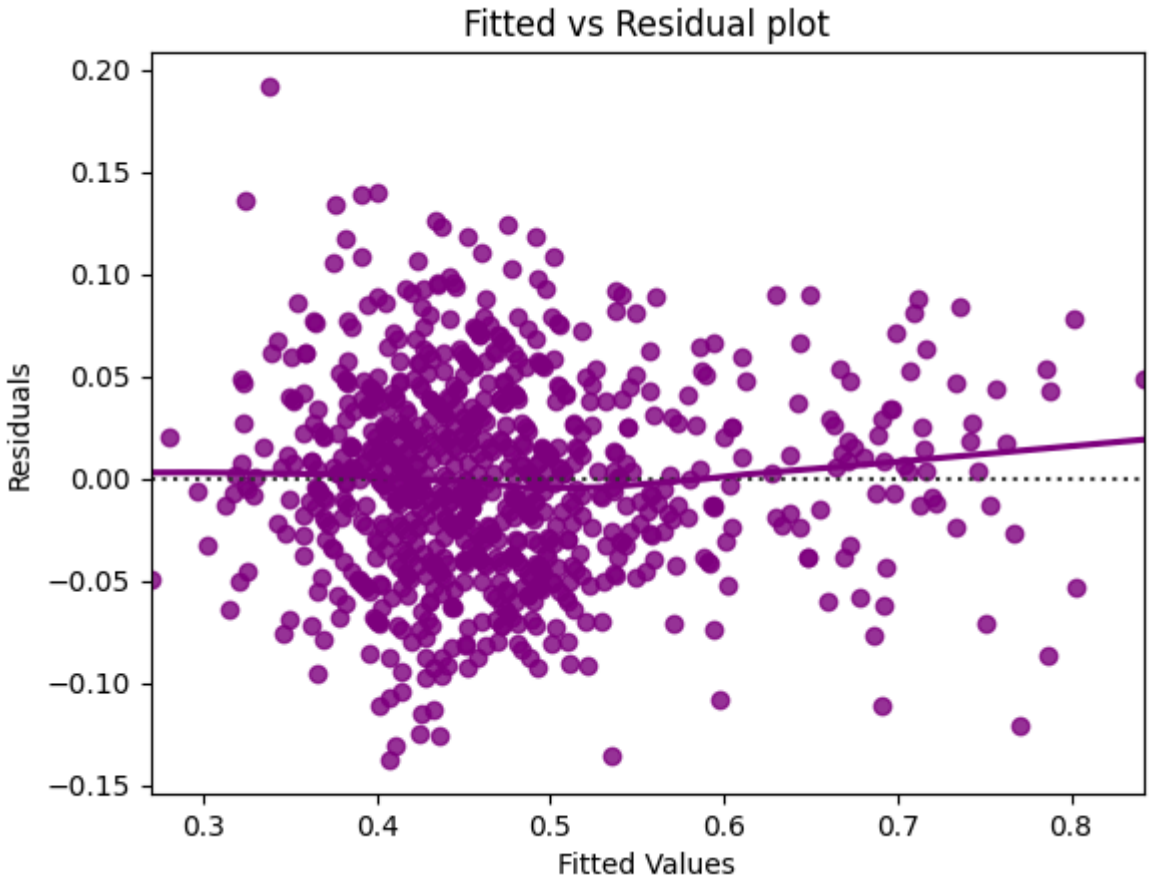


Fig.25: Fitted vs Residual plot for linear regression

Observations:

- The residuals appear randomly scattered around the horizontal zero line without a clear pattern, indicating that the **linearity assumption** holds.
- No systematic structure (e.g., curvature or funnel shape) is evident, suggesting that **independence** of errors is reasonable.
- The residual spread seems fairly constant across the range of fitted values, although a slight increase in variance is visible at higher fitted values, but not enough to indicate severe heteroscedasticity.

Conclusion: The residual plot suggests that both **linearity** and **independence** assumptions are satisfied for this model, allowing us to proceed with further diagnostics.

3. TEST FOR NORMALITY

To check whether the residuals from our regression model are normally distributed, which is an important assumption for linear regression (especially for valid hypothesis testing and confidence intervals).

Methods Used:

1. **Histogram with KDE –** Plotted the distribution of residuals with a kernel density estimate.

2. **Q–Q Plot –** Compared the quantiles of the residuals with those of a theoretical normal distribution.

3. **Shapiro–Wilk Test –** A formal statistical test for normality.

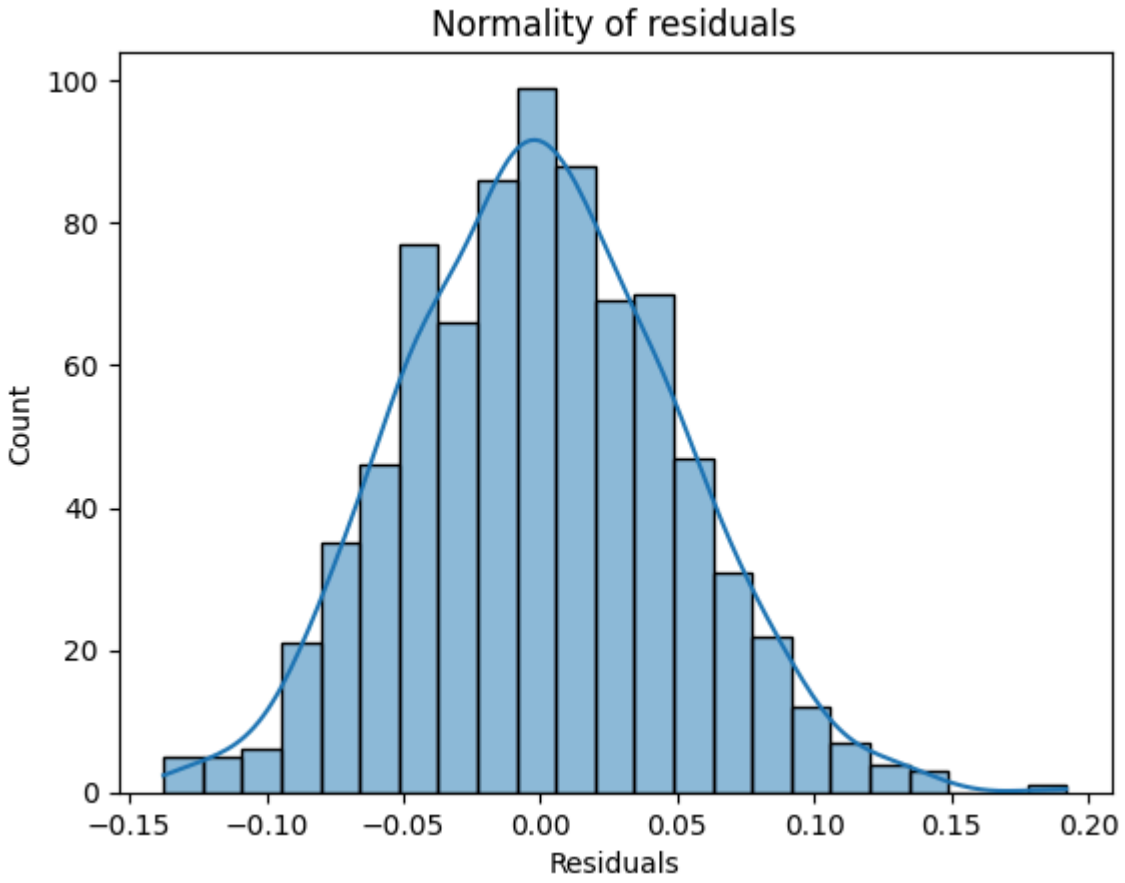


Fig.26: Normality of residuals

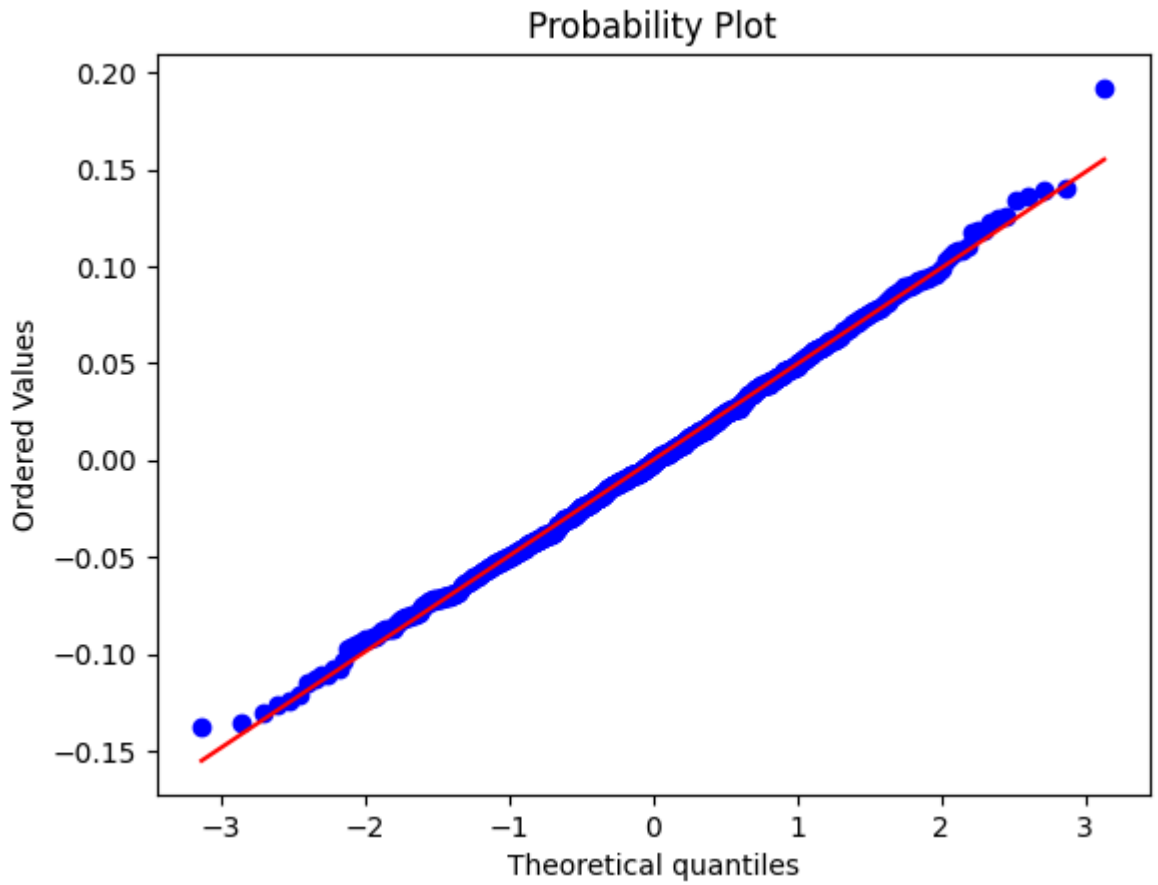


Fig.27: Q-Q Plot

```
ShapiroResult(statistic=np.float64(0.9980675192349445), pvalue=np.float64(0.5050706693244453))
```

Fig.28: Shapiro–Wilk Test

Results:

- **Histogram:** The residuals form a symmetric bell-shaped distribution centered around zero, with no extreme skewness or kurtosis.
- **Q–Q Plot:** The residuals closely follow the diagonal line, indicating that their distribution is close to normal, with only minor deviations at the tails.

- **Shapiro–Wilk Test:** Statistic = 0.9981, p-value = 0.5051. Since p > 0.05, we fail to reject the null hypothesis of normality.

Conclusion: The residuals are approximately normally distributed. This satisfies the normality assumption for the linear regression model.

4. TEST FOR HOMOSCEDASTICITY

```
[('F statistic', np.float64(1.025490628476713)),
 ('p-value', np.float64(0.4020425649699726))]
```

Fig.29: Goldfeld–Quandt test

The Goldfeld–Quandt test was performed to check whether the variance of the residuals is constant across the range of fitted values (homoscedasticity). The test yielded an F-statistic of 1.025 with a p-value of 0.402.

Since the p-value is greater than the 0.05 significance level, we fail to reject the null hypothesis of equal variances. This indicates that there is no statistical evidence of heteroscedasticity, and the assumption of homoscedasticity for linear regression is satisfied.

Model performance evaluation

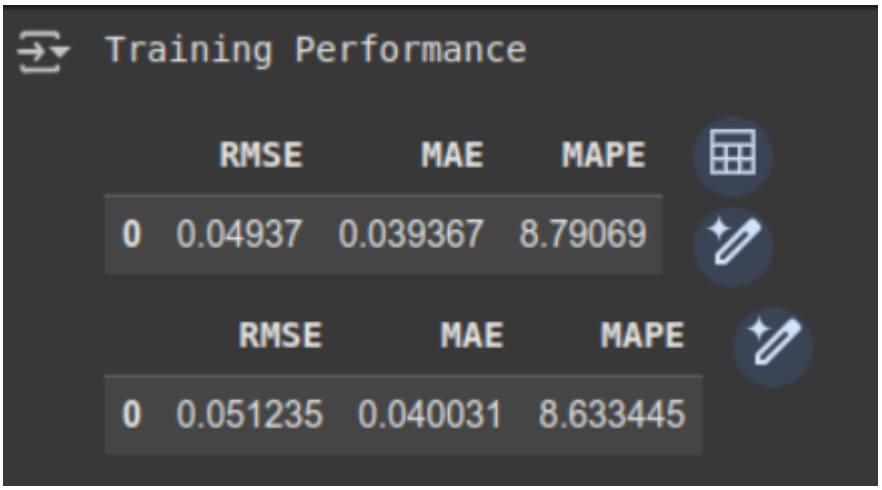


Table.6: Table showing different performance metrics on training and testing data

To assess how well the regression model predicts the target variable, we calculated three key performance metrics on both the training (80%) and testing (20%) datasets:

1. Root Mean Squared Error (RMSE): Measures the square root of the average squared differences between predicted and actual values. It penalizes larger errors more heavily, making it sensitive to outliers. Lower values indicate better fit.

2. Mean Absolute Error (MAE): Measures the average absolute difference between predicted and actual values, providing a more intuitive understanding of typical prediction error in the same units as the target variable.

3. Mean Absolute Percentage Error (MAPE): Measures the average absolute percentage difference between predicted and actual values, giving an error rate in percentage terms.

The model performance was evaluated as follows:

- *Training Set:* Performance metrics were computed using the 80% of data used for model fitting.
- *Testing Set:* Metrics were computed on the remaining 20% of unseen data to evaluate generalization ability.

Interpretation:

The training and testing errors are very close across all metrics, suggesting that the model generalizes well to unseen data and does not suffer from significant overfitting or underfitting. The consistently low error values confirm that the model provides accurate and reliable predictions for the target variable.

Actionable Insights & Recommendations

Comments on Significance of Predictors

- **Trailer Views (Highly Significant, Positive Impact):** The strongest predictor of first-day viewership. For every increase in trailer engagement, there is a substantial lift in opening-day performance. This confirms that pre-release audience interest strongly drives content success.
- **Platform Visitors (Significant, Positive Impact):** Higher weekly traffic before release directly translates into higher first-day views, reinforcing the importance of overall platform health and engagement.

- **Major Sports Events (Significant, Negative Impact)**: Releases coinciding with major sports events see an average viewership drop of ~12.5%, indicating audience diversion.
- **Day of Week:** Saturday, Sunday, and Wednesday releases significantly outperform the baseline (Friday), suggesting that both weekend leisure and mid-week drops are effective windows.
- **Seasonal Effects:** Summer and Winter releases yield higher performance compared to Autumn, likely due to seasonal leisure patterns.
- **Genre (Notably Sci-Fi):** Sci-Fi content shows a small but significant positive lift, indicating niche audience strength.

Key Takeaways for the Business

1. **Maximize Trailer Engagement:** Invest in targeted marketing to boost trailer views well before release. High-performing trailers are the clearest driver of strong opening-day performance.
2. **Optimize Release Timing:** Prioritize Saturday, Sunday, and Wednesday for key releases to leverage natural audience availability.
3. **Avoid Competing with Major Sports Events:** Where possible, schedule major content outside of sports-heavy dates to minimize audience diversion.
4. **Leverage Seasonal Trends:** Concentrate flagship releases in Summer and Winter to take advantage of natural demand spikes.
5. **Strengthen Platform Traffic:** Drive visitor growth through ongoing promotions, partnerships, and engagement campaigns, as higher pre-release traffic boosts first-day performance.

6. **Explore High-Performing Genres:** Allocate resources toward Sci-Fi and other proven high-engagement genres, and further analyze the “Others” category to identify hidden high-potential niches.